On the Complexity of Finding an Influential Variable

Lance Fortnow

November 10, 2026

Abstract

The Aaronson–Ambainis conjecture, now a theorem, says every bounded degree-d polynomial $p : \{0,1\}^n \to [0,1]$ has a variable of influence $\mathrm{poly}(\mathrm{Var}(p)/d)$. The proof is existential. We ask how hard it is to *find* such a variable. When p is given as the acceptance probability of a quantum circuit, the search problem is PromiseBQP-complete: easy for a quantum computer and PromiseBQP-hard even for degree-one polynomials. When p is given as a black box, a variable of influence $\mathrm{poly}(\mathrm{Var}(p)/d)$ can be found with $\mathrm{poly}(d/\mathrm{Var}(p)) \cdot \log n \log\log n$ queries, and $\Omega(\log n)$ queries are necessary. All of the difficulty lies in evaluating p, not in the search.

1 Background

Write $p(x) = \sum_{U \subseteq [n]} \hat{p}(U)\chi_U(x)$ in the ± 1 Fourier basis, so that

$$\mathrm{Inf}_i(p) = \sum_{U \ni i} \hat{p}(U)^2 = \tfrac{1}{4}\,\mathbb{E}_x\big[(p(x) - p(x^{\oplus i}))^2\big], \qquad \mathrm{Var}(p) = \sum_{U \neq \emptyset} \hat{p}(U)^2.$$

Aaronson and Ambainis [AA14] conjectured that every polynomial $p : \{0,1\}^n \to [0,1]$ of degree d satisfies

$$\max_i \mathrm{Inf}_i(p) \geq \mathrm{poly}(\mathrm{Var}(p)/d).$$

Turturean [Tur26] recently proved this. Blanc, Docter, Strassle, and Tan [BDST26b] give an exposition that obtains the bound $\Omega(\mathrm{Var}(p)^8/d^{12})$; the original proof gives $\Omega(\mathrm{Var}(p)^{5/2}/d^6)$.

The motivation is the simulation of quantum query algorithms. By Beals et al. [BBCMW01], the acceptance probability of a t-query quantum algorithm is a polynomial of degree at most $2t$ in the input bits. A classical simulator can query the most influential variable, restrict, and recurse until the polynomial is nearly constant. The influence bound controls the number of *queries* this simulator makes. The exposition does not discuss the cost of *finding* the influential variable. That cost appears only implicitly, through the complexity-class consequences: Aaronson and Ambainis note that a random-oracle separation of BQP from HeurP would require $\mathsf{P} \neq \mathsf{P}^{\#\mathsf{P}}$, and Blanc, Docter, Strassle, and Tan [BDST26a] strengthen this to PromiseBPP $\neq$ PromiseBQP. This bears on the question of Fortnow and Rogers [FR99] of whether BPP $\neq$ BQP relative to a random oracle.

We consider two ways p might be presented.

2 When p is given by a quantum circuit

Suppose we are given a quantum circuit Q that makes queries to an input $x \in \{0,1\}^n$, and let $p(x) = \Pr[Q^x \text{ accepts}]$.

Theorem 1 (Upper bound)**.** *There is a quantum algorithm that, given Q and $\varepsilon > 0$, runs in time* $\mathrm{poly}(|Q|, n, 1/\varepsilon)$ *and with high probability outputs i with* $\mathrm{Inf}_i(p) \geq \max_j \mathrm{Inf}_j(p) - \varepsilon$.

Proof. Since $x^{\oplus i}$ is uniform when x is,

$$\mathrm{Inf}_i(p) = \tfrac{1}{2}\Big(\mathbb{E}_x\big[p(x)^2\big] - \mathbb{E}_x\big[p(x)\,p(x^{\oplus i})\big]\Big).$$

Running Q twice independently on the same x, both runs accept with probability $p(x)^2$; running it on x and on $x^{\oplus i}$, both accept with probability $p(x)p(x^{\oplus i})$. So each term is the acceptance probability of a small quantum circuit and can be estimated to additive $\varepsilon/4$ with $O(\varepsilon^{-2}\log n)$ runs. Estimate all n influences and output the largest. □

$\mathrm{Var}(p)$ can be estimated the same way, and restricting a variable amounts to hardwiring it into Q. Hence every step of the Aaronson–Ambainis simulator, including the test of whether p is close to constant, is a PromiseBQP computation.

Theorem 2 (Lower bound)**.** *Let $0 < c < 1/4$ be a constant. Finding a variable of influence at least c, given a quantum circuit computing a degree-1 polynomial on two variables with* $\mathrm{Var}(p) \geq 1/8$*, is* PromiseBQP*-hard.*

Proof. Let C be a PromiseBQP circuit, amplified so that its acceptance probability r satisfies $r \geq 1 - 2^{-k}$ or $r \leq 2^{-k}$, with k large enough that $2^{-2k}/4 < c \leq (1-2^{-k})^2/4$. Let Q^x run C and output x_1 if C accepts and x_2 otherwise. Then $p(x) = r\,x_1 + (1-r)\,x_2$, which has degree 1,

$$\mathrm{Inf}_1(p) = \tfrac{r^2}{4}, \qquad \mathrm{Inf}_2(p) = \tfrac{(1-r)^2}{4}, \qquad \mathrm{Var}(p) = \tfrac{r^2+(1-r)^2}{4} \geq \tfrac{1}{8}.$$

In the yes case only x_1 has influence at least c; in the no case only x_2 does. The variable returned decides C. □

Since $d = 1$ and $\mathrm{Var}(p) \geq 1/8$, the target $\mathrm{poly}(\mathrm{Var}(p)/d)$ in the Aaronson–Ambainis bound is a constant, so Theorem 2 applies to exactly the guarantee the simulator needs. Finding the *most* influential variable is hard by the same reduction.

Remark 3. The circuit C in the reduction never looks at x; the hardness is an unrelativized BQP computation planted inside the polynomial. This is the "easy direction" (an unrelativized separation gives a random-oracle separation) in miniature, and the Aaronson–Ambainis theorem says it is the only obstacle: with a PromiseBQP oracle the classical simulator runs in polynomial time. The worst-case question is therefore settled. What remains open is whether finding becomes easier when hardness cannot be planted, for instance for circuits whose non-query gates are Clifford or of constant depth, or on average over natural distributions of circuits. The existence proof gives no guidance, since none of its steps is constructive.

3 When p is given as a black box

Now suppose we may query $p(x)$ for any x we choose. Everything below also works in the weaker model where a query returns a coin of bias $p(x)$, provided the same point can be queried twice.

Theorem 4. *Let $\tau = \mathrm{poly}(\mathrm{Var}(p)/d)$ be the Aaronson–Ambainis bound. There is a randomized algorithm that, with $O\big(d\,\mathrm{Var}(p)\,\tau^{-3}\log n \cdot \log(d\log n/\tau)\big) = \mathrm{poly}(d/\mathrm{Var}(p)) \cdot \log n \cdot \log\log n$ queries to p, outputs with high probability a variable i with $\mathrm{Inf}_i(p) \geq \tau/2$. Conversely, $\Omega(\log n)$ queries are necessary.*

The lower bound is immediate: for $p = x_i$ with i hidden, each query returns one bit and i is the only influential variable, so $\log_2 n$ queries are needed.

For the upper bound we use the influence of a set of variables, as in junta testing [FKRSS04, Bla09],

$$\mathrm{Inf}_S(p) = \sum_{U \cap S \neq \emptyset} \hat{p}(U)^2 = \tfrac{1}{2}\,\mathbb{E}\big[(p(x) - p(y))^2\big],$$

where x is uniform and y is x with the coordinates in S rerandomized. Two properties matter. It is monotone, so $\mathrm{Inf}_S(p) \geq \mathrm{Inf}_i(p)$ whenever $i \in S$. And it is dominated by the influence mass $W(S) = \sum_{i \in S} \mathrm{Inf}_i(p) = \sum_U |U \cap S|\,\hat{p}(U)^2$, which is additive over disjoint sets and, because p has degree d, has total

$$W([n]) = \sum_U |U|\,\hat{p}(U)^2 \leq d \cdot \mathrm{Var}(p).$$

Since $(p(x) - p(y))^2 \in [0,1]$, $\mathrm{Inf}_S(p)$ can be estimated to additive ε with confidence $1 - \gamma$ using $O(\varepsilon^{-2} \log(1/\gamma))$ queries, independent of $|S|$. With coin access, write $\mathrm{Inf}_S(p) = \mathbb{E}[p(x)^2] - \mathbb{E}[p(x)p(y)]$ and estimate each term from pairs of independent coins.

Proof of Theorem 4. Run the branch-and-prune search of Goldreich–Levin [GL89] and Kushilevitz–Mansour [KM93]. Start with the active list $\{[n]\}$. At each of $\lceil \log_2 n \rceil$ levels, split every active set in half, estimate $\mathrm{Inf}_S(p)$ for each half to additive $\tau/4$, and keep the halves whose estimate exceeds $\tau/2$. At the end, output any surviving singleton.

Assume all estimates are accurate. If $\mathrm{Inf}_{i^*}(p) \geq \tau$, every set containing i^* has $\mathrm{Inf}_S(p) \geq \tau$ by monotonicity and survives, so $\{i^*\}$ is reached. Every surviving set has $\mathrm{Inf}_S(p) > \tau/4$, and in particular every surviving singleton has influence above $\tau/4$; re-estimating the survivors to additive $\tau/8$ and outputting the best one gives influence at least $\tau/2$. The active sets at a level are disjoint, so their number is at most

$$\frac{\sum_S \mathrm{Inf}_S(p)}{\tau/4} \leq \frac{\sum_S W(S)}{\tau/4} \leq \frac{4d\,\mathrm{Var}(p)}{\tau}.$$

That gives $O(d\,\mathrm{Var}(p)\,\tau^{-1} \log n)$ estimates, each with $O(\tau^{-2} \log(d \log n/\tau))$ queries to make a union bound work. (The algorithm needs d and a lower bound on $\mathrm{Var}(p)$ to set τ; the variance can itself be estimated with $\mathrm{poly}(1/\mathrm{Var}(p))$ queries.) □

The degree bound is used only to bound the total mass $W([n])$, and hence the width of the search. This is the same argument Blanc, Docter, Strassle, and Tan [BDST26a, Prop. 10.6] use to find a variable of large *query weight*, with sums of query weights in place of set influences.

Remark 5 (Influences relative to a random oracle). In [BDST26a] the authors remark that the Aaronson–Ambainis subroutine for finding a variable of high influence is in $\mathsf{NP}^{\#\mathsf{P}}$, whereas their subroutine for query weights is in $\mathsf{PromiseBQP}$, and they cite this as an algorithmic advantage of query weights. Combining the two-copy estimator in the proof of Theorem 1 with the search above appears to put the influence version in $\mathsf{PromiseBQP}$ as well. When the input is a random oracle $O \in \{0,1\}^N$ and Q makes t queries, Inf_S for a dyadic interval S is $\mathbb{E}[p(O)^2] - \mathbb{E}[p(O)p(O')]$, a difference of average acceptance probabilities of two-copy circuits. The random oracles O and O' can be simulated with $2t$-wise independent functions [Zha12], and the search needs only $\mathrm{poly}(t) \cdot \log N$ such estimates. So the Aaronson–Ambainis regularity lemma also appears to be algorithmic under $\mathsf{PromiseBPP} = \mathsf{PromiseBQP}$. If correct, the advantage of query weights lies in their easier existence proof, not in their algorithmic tractability.

4 Related work

Aaronson and Ambainis [AA14] made their regularity lemma algorithmic under $\mathsf{P} = \mathsf{P}^{\#\mathsf{P}}$, finding influential variables by computing influences exactly. Blanc, Docter, Strassle, and Tan [BDST26a] replace influences by query weights, a property of the algorithm rather than of p. They find heavy variables with a PromiseBQP subset-sum estimator and branch-and-prune, which gives logarithmic dependence on N, and conclude that a random-oracle separation $\mathsf{PromiseBQP}^O \not\subseteq^O$ would imply $\mathsf{PromiseBPP} \neq \mathsf{PromiseBQP}$. None of these works considers hardness of finding.

Set influences and their estimation with $O(\varepsilon^{-2})$ queries, independent of $|S|$, go back to Fischer, Kindler, Ron, Safra, and Samorodnitsky [FKRSS04] and Blais [Bla09] in junta testing. Blais, Canonne, Eden, Levi, and Ron [BCELR18] introduce the influence of a random ρ-biased subset of a set, the same noise statistic one gets by flipping coordinates of S with small probability.

For learning rather than search, Eskenazis and Ivanisvili [EI22] show that bounded degree-d functions can be learned to L_2 error ε from $\log(n/\delta)\,\varepsilon^{-d-1}C^{d^{3/2}\sqrt{\log d}}$ uniformly random queries, with the $\log n$ dependence optimal. Since $|\sqrt{\mathrm{Inf}_i(h)} - \sqrt{\mathrm{Inf}_i(p)}| \leq \|h - p\|_2$, this already finds an influential variable nonadaptively with $O(\log n)$ queries, but with exponential dependence on d; Theorem 4 trades nonadaptivity for polynomial dependence on d.

On the quantum side, Fourier sampling via the Bernstein–Vazirani circuit estimates the influences of a Boolean function [FAH13, LY15]: position i of the sample is 1 with probability $\mathrm{Inf}_i(f)$.

5 Conclusion

Combining the two settings, searching for an influential variable costs only $O(\log n \log\log n) \cdot \mathrm{poly}(d/\mathrm{Var}(p))$ evaluations of p, while evaluating p from a quantum circuit is PromiseBQP-hard even in the simplest case. The barrier in the Blanc–Docter–Strassle–Tan result sits at PromiseBPP versus PromiseBQP because of evaluation, not search.

AI disclosure

This note was produced in a single conversation with Claude Opus 5.5, a large language model developed by Anthropic, on November 10, 2026. The author posed the questions: whether the Blanc–Docter–Strassle–Tan exposition addresses the complexity of finding an influential variable, and how hard that problem is when p is given by a quantum circuit and when it is given as a black box. Claude proposed the theorems and proofs in Sections 2 and 3, carried out the literature search behind Section 4 (including the observation in Remark 5), and wrote the text and LaTeX. The literature search led to the simpler branch-and-prune proof of Theorem 4, which replaced a more complicated argument in an earlier draft. The arguments have not been independently verified, and some bibliographic details were supplied from the model's own knowledge rather than checked against the sources.

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